

Exact linearized Coulomb collision operator in the moment expansion

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In the moment expansion, the Rosenbluth potentials, the linearized Coulomb collision operators, and the moments of the collision operators are analytically calculated for any moment. The explicit calculation of Rosenbluth potentials converts the integro-differential form of the Coulomb collision operator into a differential operator, which enables one to express the collision operator in a simple closed form for any arbitrary mass and temperature ratios. In addition, it is shown that gyrophase averaging the collision operator acting on arbitrary distribution functions is the same as the collision operator acting on the corresponding gyrophase averaged distribution functions. The moments of the collision operator are linear combinations of the fluid moments with collision coefficients parametrized by mass and temperature ratios. Useful forms involving the small mass-ratio approximation are easily found since the collision operators and their moments are expressed in terms of the mass ratio. As an application, the general moment equations are explicitly written and the higher order heat flux equation is derived. © 2006 American Institute of Physics.

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I. INTRODUCTION

The state of an ionized gas for plasma species a is described by a distribution function $f_a(t, \mathbf{x}, \mathbf{v})$ and its evolution is governed by the Boltzmann equation

$$\partial_t f_a + \mathbf{v} \cdot \nabla f_a + \frac{e_a}{m_a} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \partial_{\mathbf{v}} f_a = \sum_b C(f_a, f_b), \quad (1)$$

where we assume that binary Coulomb collisions, $\sum_b C(f_a, f_b)$, are the dominant collisional process. Predictive kinetic and fluid models of plasmas require accurate and explicit specification of the collision operator and the collisional moments. In this paper we use moment expansions for the distribution functions to derive analytical forms for the linearized Fokker-Planck operator and its moments.

For a pure plasma, the electron-ion collision operators can be simplified by the small-mass ratio approximation.^{1,2} However in a multispecies plasma the presence of multiple ion species requires collision operators valid for general mass ratio. For the general masses and temperatures, low-order collisional moments of self-adjoint collision operator are calculated in Refs. 3 and 4. However the results are not exact since the linearized collision operator is only approximately self-adjoint for the different masses and temperatures.^{5,6} Note that the generating formula, Eq. (A2) of Ref. 3, is different from Eq. (A.4) of Ref. 1, missing the term proportional to $(1 - T_a/T_b)$. Though Braginskii derived the collisional moments for like species and small-mass ratio approximation, one may obtain exact results for momentum moments by calculating proper coefficients from Eq. (A.4) of

Ref. 1. However there are no explicit exact expressions of collisional moments even for low-order moments, let alone high-order moments.

Since the linearized theory is limited to small perturbations from a Maxwellian distribution for highly collisional plasmas, one might think that the high-order moments are not necessary. However it is pointed out that in magnetically confined fusion plasmas a Maxwellian distribution is pertinent even when the mean-free-path is long.⁷ In this long mean-free-path regime it is well known that high-order moments are required for convergence in calculating general integral heat flux with the pitch-angle scattering operator.⁸ Therefore high-order moments are required for improvements to the general closures with full linearized collision operators.

Thus, the motivations of this work are (a) to provide exact expression of linearized collision operators and collisional moments for arbitrary moments in the case of general mass and temperature ratios and (b) to write general moment equations. These expressions will be useful (i) in classical or neoclassical collisional transport theories for multiple ion species, (ii) in deriving general closures valid in plasmas with regions of low and high collisionality, and (iii) in calculating higher perturbation orders of collisional transport theory to quantitatively study limitations of Braginskii closures. As a by-product, explicit forms of Rosenbluth potentials will enable one (iv) to analytically calculate nonlinear collision operators.

The Boltzmann collision integrals (the collisional moments) for some lowest moments are well evaluated by means of partial bracket integrals in Refs. 9 and 10 for the same temperature of unlike species. The generalization to different temperatures are well summarized in Ref. 11. In the Fokker-Planck operator approach, the Landau form of the Coulomb collision integral has an especially compact form¹²:

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$$C(f_a, f_b) = \frac{\gamma_{ab}}{2} \partial_{\mathbf{v}} \cdot \int d\mathbf{v}' \mathbf{U} \cdot \left(\frac{\partial_{\mathbf{v}} f_a}{m_a} f_b - f_a \frac{\partial_{\mathbf{v}'} f_b}{m_b} \right), \quad (2)$$

where $f_a = f_a(\mathbf{v})$ and $f_b = f_b(\mathbf{v}')$ in the integral, $\mathbf{U} = \frac{1}{u^3} (u^2 \mathbf{I} - \mathbf{u}\mathbf{u})$ with $\mathbf{u} = \mathbf{v} - \mathbf{v}'$ and $\gamma_{ab} = \frac{e_a^2 e_b^2 \ln \Lambda_{ab}}{4\pi\epsilon_0^2 m_a}$ with $\Lambda_{ab} = \frac{12\pi\epsilon_0 m_a T_b + m_b T_a}{|e_a e_b| m_a + m_b} \lambda_D$.¹¹ Corrections due to the quantum effects are necessary for temperatures higher than $T = 36.4$ eV.¹³ The Debye length for a multicomponent plasma is given by $\lambda_D^{-2} = \sum_a \frac{n_a e_a^2}{\epsilon_0 T_a}$.¹⁴ In stellar dynamics, Chandrasekhar calculated the force of dynamical friction and the diffusion tensor for gravitational interactions in the case of a Maxwellian distribution.¹⁵

An elegant expression of Coulomb collision operator is given by introducing Rosenbluth potentials^{16,17}:

$$C_{ab} = \frac{\gamma_{ab}}{2m_a} \partial_{\mathbf{v}} \cdot \left[\partial_{\mathbf{v}} \cdot (f_a \partial_{\mathbf{v}} \partial_{\mathbf{v}'} G_b) - 2 \left(1 + \frac{m_a}{m_b} \right) f_a \partial_{\mathbf{v}} H_b \right] \quad (3)$$

where we adopt the same definitions of Rosenbluth potentials as Ref. 18

$$H_b(\mathbf{v}) = \int d\mathbf{v}' f_b(\mathbf{v}') \frac{1}{|\mathbf{v} - \mathbf{v}'|} \quad (4)$$

and

$$G_b(\mathbf{v}) = \int d\mathbf{v}' f_b(\mathbf{v}') |\mathbf{v} - \mathbf{v}'|. \quad (5)$$

In such a form the collision operator is known as the Rosenbluth, MacDonald, Judd (RMJ) operator.¹⁶ Though both the Landau and RMJ operators are integro-differential operators, the latter can be converted to a differential operator when the Rosenbluth potentials are explicitly provided.

In this paper we provide explicit formulas of exact linearized collision operators and collisional moments for any arbitrary moments. Not only do the exact formulas include the previous work for low-order moments in Refs. 1, but they also enable one to obtain the collision operators and collisional moments for any arbitrary high-order moments. First we explicitly calculate the Rosenbluth potentials in the moment expansion,^{19,20} which enables us to write the collision operators in a form that involves simple differentiations. Then we calculate the collisional moments by integrating the product of velocity polynomials and the collision operator. The linearized collision operator and its moments are linear combinations of fluid moments of the distribution function. With the explicit formulas of the collision operators and the collisional moments given, we can show the conservation laws for the particle number, the energy, and the momentum from the corresponding collisional moments. Each moment term of the linearized collision operators in the moment expansion preserves the conservation laws.

In the collisional transport theory, the Hilbert²¹ and Chapman-Enskog techniques²² provide the solutions for the Boltzmann equation in the presence of small parameter ϵ , the ratio of the mean free path to a typical macroscopic length scale. The original Chapman-Enskog procedure was simplified considerably through the Sonine polynomial expansion by Burnett.²³ Grad's moment expansion provides the system-

atic expansion of the explicit solution of Boltzmann equation (For reviews, see Refs. 9–11 and 24). With the collisional moments calculated, we also present the general moment equations and introduce a simple procedure to find the collisional solutions not from the kinetic equation but directly from the moment equations. The ability of our formula to calculate any higher moment collision operator enables us to find, as an example, the higher ϵ -order solutions of parallel heat flux with a satisfactory convergence. Comparing the higher order solution with the Braginskii equations (2), we quantitatively estimate the range of validity for the heat flux and show that the diffusive closure is valid only for extremely collisional plasmas.

This paper is organized as follows: In Sec. II we introduce the moment expansion of the distribution function in terms of irreducible tensorial Hermite polynomials and Sonine polynomials. Then we calculate the Rosenbluth potentials and write the linearized collision operators in explicit forms. We also discuss the properties of the gyroaveraged collision operator. In Sec. III we calculate collisional moments and discuss conservation laws. Since collisional moments can be expressed in terms of moments of the distribution function, we can write any arbitrary number of moment equations explicitly. We also introduce the procedure to find the solution of the moment equation in the collisional plasma, as an example, through the ion heat flux equation. Therein we discuss the physical meaning of the higher ϵ -order solution and the limitation of the Braginskii equation for the diffusive heat flux. In Sec. V we discuss future applications such as transport equations for multi ion species, the derivation of the general closure, and the effect of nonlinear collision operators. In the Appendices we show the speed-dependent collision frequency functions in the collision operator and collisional coefficients in the collisional moments for some low orders of moments. We also include useful formulas in the small mass-ratio approximation.

II. EXACT CALCULATION OF COLLISION OPERATORS IN THE MOMENT EXPANSION

A. Moment expansion of the distribution function

In the moment methods of solving the kinetic equation, the distribution function $f(t, \mathbf{x}, \mathbf{v})$ is expanded in terms of orthogonal tensorial polynomials of the particle velocities. Grad suggested the expansion in terms of (reducible) tensorial Hermite polynomials.¹⁹ The equivalent expansion in terms of irreducible Hermite polynomials²⁰ gives the relation to Chapman-Enskog theory.⁹ The fluid equations can be obtained very easily from the orthogonal relations of the tensorial polynomials.

In this section we introduce Grad's moment expansion in the irreducible polynomials and define our notations and conventions. Let us define a $(l+2k)$ th moment of distribution function f_a for species a by

$$n_a M_a^{lk}(t, \mathbf{x}) = \int d\mathbf{v} P^l(\mathbf{v}) v_{Ta}^{2k} L_k^{(l+1/2)}(s_a^2) f_a(t, \mathbf{x}, \mathbf{v}) \quad (6)$$

where $\mathbf{s}_a = \mathbf{v}/v_{Ta}$ and the thermal speed $v_{Ta} = (2T_a/m)^{1/2}$. Note that $n_a(t, \mathbf{x})$ and $T_a(t, \mathbf{x})$ refer to full density and full tempera-

ture throughout this paper unless otherwise stated (as only in the discussion of H -theorem in Sec. IV B). Several low-order moments corresponding to density, energy, fluid velocity, heat flux, viscous stress tensor, and heat stress tensor are as follows:

$$M_a^{00} = 1, \quad M_a^{01} = -V_a^2, \quad (7a)$$

$$M_a^{10} = \mathbf{V}_a, \quad M_a^{11} = \frac{2}{m_a n_a} \left(\frac{5}{2} n_a T_a \mathbf{V}_a - \mathbf{Q}_a \right), \quad (7b)$$

$$M_a^{20} = \frac{\Pi_a}{m_a n_a}, \quad M_a^{21} = \frac{2}{m_a n_a} \left(\frac{7}{2} n_a T_a \Pi_a - \Theta_a \right). \quad (7c)$$

Now we introduce dimensionless fluid moments $\hat{M}^{lk} = v_T^{-(l+2k)} M^{lk}$ as expansion coefficients in the moment expansion:

$$f_a = \sum_{lk} \frac{1}{\sigma_k^l} f_a^{lk} \quad (8)$$

where the normalization constant σ_k^l is defined below and

$$f_a^{lk} = f_a^{(0)} P_a^{lk} \cdot \hat{M}_a^{lk}. \quad (9)$$

Here the Maxwellian distribution for a local thermal equilibrium with $n_a(t, \mathbf{x})$ and $T_a(t, \mathbf{x})$ is

$$f_a^{(0)} = \frac{n_a}{\pi^{3/2} v_T^3} e^{-s_a^2} \quad (10)$$

and the irreducible tensorial Hermite polynomials $\mathbf{P}^{lk}$ are the products of Sonine polynomials (associated Laguerre polynomials) $L_k^{(l+1/2)}$ and the irreducible harmonic polynomials $\mathbf{P}^l$:

$$\mathbf{P}_a^{lk} = \mathbf{P}^l(\mathbf{s}_a) L_k^{(l+1/2)}(s_a^2). \quad (11)$$

Note that $\hat{M}^{lk}(t, \mathbf{x})$ is independent of the velocity variable $\mathbf{v}$ and the velocity dependence of a distribution function comes only through $f_a^{(0)} \mathbf{P}_a^{lk}$.

The Sonine polynomials $L_k^{(l+1/2)}$ are given by

$$L_k^{(l+1/2)}(x) = \sum_{m=0}^k c_{km}^l x^m, \quad (12)$$

with

$$c_{km}^l = \frac{(-1)^m (l+k+1/2)!}{(k-m)! (l+m+1/2)! m!}. \quad (13)$$

The rank- l tensors $\mathbf{P}^l$ can be obtained from the recurrence relation

$$\mathbf{P}^{l+1}(\mathbf{v}) = \mathbf{v} \mathbf{P}^l(\mathbf{v}) - \frac{v^2}{2l+1} \partial_{\mathbf{v}} \mathbf{P}^l(\mathbf{v}) \quad (14)$$

with $\mathbf{P}^0 = 1$. Note that $\mathbf{v} \mathbf{P}^l$ will not be summed over all possible permutations and is different by a factor $1/(l+1)$ from Grad's notations.²⁰ To avoid any possible confusion, we rewrite Eq. (14) explicitly in component form

$$P_{\alpha\beta\cdots\gamma}^{l+1}(\mathbf{v}) = v_{\alpha} P_{\beta\cdots\gamma}^l - \frac{v^2}{2l+1} \partial_{\alpha} P_{\beta\cdots\gamma}^l(\mathbf{v}) \quad (15)$$

where $\partial_{\alpha} = \partial/\partial v_{\alpha}$ and the Greek letters $\alpha, \beta, \dots$ denote spatial components. From the component form, we can easily verify that $\mathbf{P}^l (l \geq 2)$ are symmetric and traceless for any pair of indices. Consequently, the fluid moments $\mathbf{M}^{lk} (l \geq 2)$ are defined to be symmetric and traceless. (The moments for $l=0$ are scalars and $l=1$ moments are vectors.) A few low orders related to moments of a density, a velocity, and a stress are

$$\mathbf{P}^0(\mathbf{v}) = 1, \quad \mathbf{P}^1(\mathbf{v}) = \mathbf{v}, \quad \text{and} \quad \mathbf{P}^2(\mathbf{v}) = \mathbf{v}\mathbf{v} - \frac{1}{3}v^2 \mathbf{I}, \quad (16)$$

respectively.

The normalization constants in $\sigma_k^l = \sigma_l \lambda_k^l$,

$$\sigma_l = \frac{l! (1/2)!}{2^l (l+1/2)!} \quad \text{and} \quad \lambda_k^l = \frac{(l+k+1/2)!}{k! (1/2)!}, \quad (17)$$

are introduced from

$$\int d\mathbf{v} \mathbf{P}^n(\mathbf{v}) \mathbf{P}^l(\mathbf{v}) \cdot \mathbf{M}^l g(v) = \mathbf{M}^n \delta_{nl} \sigma_n \int d\mathbf{v} v^{2n} g(v) \quad (18)$$

and

$$\int d\mathbf{v} f_a^{(0)} s_a^{2l} L_p^{(l+1/2)}(s_a^2) L_q^{(l+1/2)}(s_a^2) = n_a \lambda_p^l \delta_{pq}, \quad (19)$$

respectively. Finally, the integration of $\mathbf{P}^l$ over an azimuthal angle $\phi_{\hat{\mathbf{n}}}$ along a unit vector $\hat{\mathbf{n}}$ gives the following useful formula:

$$\int_0^{2\pi} d\phi_{\hat{\mathbf{n}}} \mathbf{P}^l(\mathbf{v}) = 2\pi v^l P_l(\cos \theta) \mathbf{P}^l(\hat{\mathbf{n}}) \quad (20)$$

where P_l is a Legendre polynomial.

B. Rosenbluth potentials

Having introduced the moment expansion in the previous section, to calculate collision integrals (3) we first find the Rosenbluth potentials:

$$G^{lk}(\mathbf{v}) = \int d\mathbf{v}' f^{lk}(\mathbf{v}') |\mathbf{v} - \mathbf{v}'|. \quad (21)$$

Since the Sonine polynomials $L_k^{(l+1/2)}(s^2)$ are expanded in even powers of the dimensionless speed variable, it is convenient to calculate the Rosenbluth potentials associated with a monomial s^{2m} instead of $L_k^{(l+1/2)}(s^2)$. The corresponding Rosenbluth potential is

$$\mathbf{G}_*^{lm}(\mathbf{v}) = \int d\mathbf{v}' f^{(0)}(v') \mathbf{P}^l(\mathbf{s}') s'^{2m} |\mathbf{v} - \mathbf{v}'| \quad (22)$$

where the subscript $*$ indicates that the superscript m denotes the power of the speed variable s^2 of the monomial instead of the order of the Sonine polynomial. Then, from (12), we can write the Rosenbluth potential corresponding to the moment $\mathbf{M}^{lk}$ as

$$G^{lk} = \hat{M}^{lk} \cdot \sum_{m=0}^k c_{km}^l \mathbf{G}_*^{lm}. \quad (23)$$

After the integration over the azimuthal angle $\phi_{\hat{\mathbf{v}}}$ in Eq. (22) with the help of Eq. (20), we write

$$\mathbf{G}_*^{lm}(\mathbf{v}) = \frac{2nv_T}{\sqrt{\pi}} \mathbf{P}^l(\hat{\mathbf{v}}) \int ds' s'^{l+2} e^{-s'^2} s'^{2m} \times \int_{-1}^1 d\xi' \sqrt{s^2 + s'^2 - 2ss'\xi'} P_l(\xi') \quad (24)$$

where ξ' is the cosine angle between $\mathbf{v}$ and $\mathbf{v}'$. Using the recurrence relation $P_l = (P'_{l+1} - P'_{l-1})/(2l+1)$ and integrating by parts, we perform ξ' -integration

$$\int_{-1}^1 d\xi' \sqrt{s^2 + s'^2 - 2ss'\xi'} P_l(\xi') = \frac{2}{2l+1} \left(\frac{1}{2l+3} \frac{s_{<}^{l+2}}{s_{>}^{l+1}} - \frac{1}{2l-1} \frac{s_{<}^l}{s_{>}^{l-1}} \right) \quad (25)$$

where $s_{>}$ and $s_{<}$ are, respectively, the larger and smaller value in s and s' . Here we have used the expansion property of Legendre polynomials

$$\frac{1}{\sqrt{s^2 + s'^2 - 2ss'\xi'}} = \sum_{n=0}^{\infty} \frac{s_{<}^n}{s_{>}^{n+1}} P_n(\xi'), \quad (26)$$

the orthogonality relation $\int_{-1}^1 d\xi P_l(\xi) P_n(\xi) dx = \frac{2}{2l+1} \delta_{ln}$, and $P_l(1)=1$, $P_l(-1)=(-1)^l$.

For the integration over speed variable, we define, respectively, the low (+) and high (−) speed integrals as

$$I_+^k = \frac{2}{\sqrt{\pi}} \int_0^s ds' s'^k e^{-s'^2}, \quad (27a)$$

$$I_-^k = \frac{2}{\sqrt{\pi}} \int_s^\infty ds' s'^k e^{-s'^2}. \quad (27b)$$

Integration by parts gives the following recurrence relations:

$$I_{\pm}^k = \mp s^{k-1} \frac{E'}{2} + \frac{k-1}{2} I_{\pm}^{k-2} \quad (28)$$

with the initial values

$$I_+^0 = E = \frac{2}{\sqrt{\pi}} \int_0^s ds' e^{-s'^2}, \quad I_-^1 = \frac{E'}{2} = \frac{1}{\sqrt{\pi}} e^{-s^2}. \quad (29)$$

In the calculations of the Rosenbluth potentials, we will need even orders of I_+ and odd orders of I_- only. From Eqs. (28) and (29) we can derive

$$I_+^{2k} = \frac{(k-1/2)!}{(1/2)!} \frac{E}{2} - \sum_{j=0}^{k-1} \frac{(k-1/2)!}{(j+1/2)!} s^{2j+1} \frac{E'}{2} \quad (30)$$

and

$$I_-^{2k+1} = \sum_{j=0}^k \frac{k!}{j!} s^{2j} \frac{E'}{2}. \quad (31)$$

Now combining all the above calculations, we have

$$\mathbf{G}_*^{lm} = 2nv_T \frac{\mathbf{P}^l(\hat{\mathbf{v}})}{(2l+1)} \left[\frac{1}{2l+3} \left(\frac{I_+^{2l+2m+4}}{s^{l+1}} + s^{l+2} I_-^{2m+1} \right) - \frac{1}{2l-1} \left(\frac{I_+^{2l+2m+2}}{s^{l-1}} + s^l I_-^{2m+3} \right) \right]. \quad (32)$$

Note that the potential is expressed as polynomials of speed variables multiplied by E and E' . Similarly we can calculate

$$\mathbf{H}_*^{lm}(\mathbf{v}) = \int d^3v' f^{(0)}(v') \mathbf{P}^l(\hat{\mathbf{v}}') s'^{2m} \frac{1}{|\mathbf{v} - \mathbf{v}'|} \quad (33)$$

using the expansion (26),

$$\mathbf{H}_*^{lm} = \frac{2n}{v_T} \frac{\mathbf{P}^l(\hat{\mathbf{v}})}{(2l+1)} \left(\frac{I_+^{2l+2m+2}}{s^{l+1}} + s^l I_-^{2m+3} \right), \quad (34)$$

from which we can find

$$\mathbf{H}^{lk} = \hat{\mathbf{M}}^{lk} \cdot \sum_{m=0}^k c_{km}^l \mathbf{H}_*^{lm}. \quad (35)$$

One can also get the same result using

$$\mathbf{H} = \frac{1}{2} \partial_{\mathbf{v}}^2 \mathbf{G}, \quad (36)$$

where

$$\partial_{\mathbf{v}}^2 = \frac{1}{v^2} \frac{\partial}{\partial v} v^2 \frac{\partial}{\partial v} - \frac{1}{v^2} L^2 \quad (37)$$

with $L^2 = -(\mathbf{v} \times \partial_{\mathbf{v}}) \cdot (\mathbf{v} \times \partial_{\mathbf{v}})$.

Finally we write some derivatives of Rosenbluth potentials that will be used in the calculation of the collision operators (43):

$$\mathbf{G}_{*,s,s}^{lm} = 2nv_T \frac{\mathbf{P}^l(\hat{\mathbf{v}})}{(2l+1)} \times \left[\frac{(l+1)(l+2)}{2l+3} \left(\frac{I_+^{2l+2m+4}}{s^{l+3}} + s^l I_-^{2m+1} \right) - \frac{l(l-1)}{2l-1} \left(\frac{I_+^{2l+2m+2}}{s^{l+1}} + s^{l-2} I_-^{2m+3} \right) \right] \quad (38)$$

and

$$\mathbf{H}_{*,s}^{lm} = \frac{2n}{v_T} \frac{\mathbf{P}^l(\hat{\mathbf{v}})}{(2l+1)} \left(- (l+1) \frac{I_+^{2(l+m+1)}}{s^{l+2}} + l s^{l-1} I_-^{2m+1} \right). \quad (39)$$

C. Linearized collision operators

With the Rosenbluth potentials expressed in analytic forms, the calculation of the collision operator from Eq. (3) is straightforward. Using $\nabla^2 \mathbf{G} = 2\mathbf{H}$ and $\nabla^2 \mathbf{H} = -4\pi \mathbf{f}$ we can remove higher order derivatives to further simplify Eq. (3)

$$C_{ab} = \frac{\gamma_{ab}}{m_a} \left(\frac{1}{2} \partial_{\mathbf{v}} \partial_{\mathbf{v}'} f_a \cdot \partial_{\mathbf{v}} \partial_{\mathbf{v}'} G_b + \left(1 - \frac{m_a}{m_b} \right) \partial_{\mathbf{v}} f_a \cdot \partial_{\mathbf{v}} H_b + 4\pi \frac{m_a}{m_b} f_a f_b \right). \quad (40)$$

For the distribution function (8) in the moment expansion, bilinearity of the collision operator implies that

$$C(f_a, f_b) = C_{ab}^{(0)} + C_{ab}^{(1)} + C_{ab}^{(2)}, \quad (41)$$

where

$$C_{ab}^{(0)} = C(f_a^{(0)}, f_b^{(0)}), \quad (42a)$$

$$C_{ab}^{(1)} = \sum_{lk} \frac{1}{\sigma_k} [C_{ab}^{(lk,0)} + C_{ab}^{(0,lk)}], \quad (42b)$$

$$C_{ab}^{(2)} = \sum_{lk} \sum_{np} \frac{1}{\sigma_k \sigma_p} C(f_a^{lk}, f_b^{np}), \quad (42c)$$

where the summations exclude the M^{00} moment term. When the system is not far from equilibrium, we may neglect the nonlinear term $C_{ab}^{(2)}$ and consider a linearized collision operator. The distribution function and the corresponding Rosenbluth potentials $f_b^{(0)}$, $H_b^{(0)}$ and $G_b^{(0)}$ in the test-particle operator $C_{ab}^{(lk,0)} = C(f_a^{lk}, f_b^{(0)})$, and the distribution function $f_a^{(0)}$ in the field operator $C_{ab}^{(0,lk)} = C(f_a^{(0)}, f_b^{lk})$ are functions of speed, v , only. This significantly simplifies the calculation of linearized collision operators in the (v, ξ, γ) , or equivalently in the spherical polar coordinate system. The test-particle operator and the field operator, respectively, are calculated from

$$C_{ab}^{(lk,0)} = \frac{\nu_{ab} v_{Ta}^3}{n_b} \left(-\frac{1}{2} L^2 f_a^{lk} \frac{G_{b,v}^{(0)}}{v^3} + \frac{1}{2} f_a^{lk} G_{b,v,v}^{(0)} + f_a^{lk} \times \left[\frac{G_{b,v}^{(0)}}{v^2} + \left(1 - \frac{m_a}{m_b} \right) H_{b,v}^{(0)} \right] + 4\pi \frac{m_a}{m_b} f_a^{lk} f_b^{(0)} \right), \quad (43a)$$

$$C_{ab}^{(0,lk)} = \frac{2\nu_{ab} v_{Ta} f_a^{(0)}}{n_b} \left(-H_b^{lk} - \left(1 - \frac{m_a}{m_b} \right) v H_{b,v}^{lk} + \frac{v^2}{v_{Ta}} G_{b,v,v}^{lk} + 2\pi v_{Ta}^2 \frac{m_a}{m_b} f_b^{lk} \right), \quad (43b)$$

where $G_b^{(0)} = n_b v_{Tb} \left[\left(\frac{1}{2} + s_b^2 \right) \frac{E_b}{s_b} + \frac{1}{2} E_b' \right]$, $H_b^{(0)} = \frac{n_b}{v_{Tb}} \frac{E_b}{s_b}$, and $\nu_{ab} = \frac{\gamma_{ab} n_b}{m_a v_{Ta}^3}$. Note that the derivatives with respect to angular variables appear in the expression $L^2 f_a^{lk}$ only, which is simply $l(l+1) f_a^{lk}$ from $L^2 \mathbf{P}^l(\hat{\mathbf{v}}) = l(l+1) \mathbf{P}^l(\hat{\mathbf{v}})$.

After some algebra on (43) using (23) with (32), (38), and (35) with (34) and (39), we can write

$$C_{ab}^{(lk,0)}(\mathbf{v}) = f_a^{(0)} \mathbf{P}^l(\hat{\mathbf{v}}) \cdot \hat{\mathbf{M}}_a^{lk} \nu_{ab}^{(lk,0)}, \quad (44a)$$

$$C_{ab}^{(0,lk)}(\mathbf{v}) = f_a^{(0)} \mathbf{P}^l(\hat{\mathbf{v}}) \cdot \hat{\mathbf{M}}_b^{lk} \nu_{ab}^{(0,lk)}, \quad (44b)$$

where the speed functions associated with $\mathbf{P}^l L_k^{(l+1/2)}$ are

$$\nu_{ab}^{(lk,0)} = \sum_{m=0}^k c_{km}^l \nu_{ab}^{(lm,0)}, \quad (45a)$$

$$\nu_{ab}^{(0,lk)} = \sum_{m=0}^k c_{km}^l \nu_{ab}^{(0,lm)}. \quad (45b)$$

The speed functions for $\mathbf{P}^l s^{2m}$ are

$$\nu_{ab}^{(lm,0)} = \frac{\nu_{ab}}{\chi} s_a^{l+2m} \sum_{n=0}^2 \left(\frac{a_{Enab}^{lm}}{s_a^{2n}} \frac{E_b}{s_b} + \frac{a_{enab}^{lm}}{s_a^{2n}} E_b' \right) \quad (46)$$

with

$$a_{E0ab}^{lm} = -\frac{2}{\mu} (1 - \theta), \quad (47a)$$

$$a_{E1ab}^{lm} = -\frac{1}{2} l(l+1) + \frac{1}{\mu} (l+2m)(1-2\theta), \quad (47b)$$

$$a_{E2ab}^{lm} = \frac{\theta}{\mu} \left[\frac{3}{4} l(l-1) + 2m(l+m-1) \right], \quad (47c)$$

$$a_{e0ab}^{lm} = 2(1-\theta) \left(\frac{1}{\theta} + \frac{1}{\mu} \right), \quad (47d)$$

$$a_{e1ab}^{lm} = (l+2m) \left(1 - \frac{1}{\mu} + \frac{2\theta}{\mu} \right), \quad (47e)$$

$$a_{e2ab}^{lm} = -\frac{\theta}{\mu} \left[\frac{3}{4} l(l-1) + 2m(l+m-1) \right] \quad (47f)$$

and

$$\nu_{ab}^{(0,lm)} = \nu_{ab} \chi \sum_{n=0}^1 \left(b_{enab}^{lm} s_b^{l+2m+2n} E_b' + b_{+nab}^{lm} \frac{I_{+b}^{2(l+m+1)}}{s_b^{l+1-2n}} + b_{-nab}^{lm} s_b^{l+2n} I_{-b}^{2m+1} \right) \quad (48)$$

with

$$b_{e0ab}^{lm} = \frac{2\mu}{\theta^2}, \quad (49a)$$

$$b_{e1ab}^{lm} = -\frac{4(l^2 + l - 1)}{(2l-1)(2l+3)}, \quad (49b)$$

$$b_{+0ab}^{lm} = \frac{2(l+1)(l+2)(2l+2m+3)}{(2l+1)(2l+3)} + \frac{4(l\mu - l - 1)}{(2l+1)\theta}, \quad (49c)$$

$$b_{+1ab}^{lm} = -\frac{4l(l-1)}{(2l-1)(2l+1)}, \quad (49d)$$

$$b_{-0ab}^{lm} = -\frac{4l(l-1)(m+1)}{(2l-1)(2l+1)} - \frac{4(l\mu + \mu - l)}{(2l+1)\theta}, \quad (49e)$$

$$b_{-1ab}^{lm} = \frac{4(l+1)(l+2)}{(2l+1)(2l+3)}. \quad (49f)$$

Here we have introduced three parameters of mass ratio, temperature ratio, and thermal speed ratio between species a and b as:

$$\mu = \frac{m_b}{m_a}, \quad \theta = \frac{T_b}{T_a}, \quad \chi = \frac{v_{Tb}}{v_{Ta}} = \sqrt{\frac{\theta}{\mu}}. \quad (50)$$

Since we have written the collision operator in a closed analytical form we can compare Eq. (45) with the collision operator Eq. (15) in Ref. 25. As pointed out in Ref. 25, since the Sonine polynomials $L_k^{(l+1/2)}$ are not the eigenfunctions of the collision operator, ν_{ab}^{lk} is, in general, comprised of an infinite sum of Sonine polynomials $L_k^{(l+1/2)}(s_a^2)$ because of $E_b = \text{erf}(s_b)$ and $E'_b = \text{erf}'(s_b)$ terms in Eq. (45). However one should note that this does not spoil the conservation laws as we will show in the next section.

For the collisions between electron and ion species with comparable temperatures, the collision operators can be written quite simply by taking into account the smallness of the ratio of the electron mass to the ion mass. Since collision speed functions (48) and (46) are expressed explicitly in terms of the mass ratio μ , it is trivial to write the leading order terms of the collision operators C_{ei} and C_{ie} . The approximate collision operators for a general moment are given in Appendix C. They agree well with the results of Ref. 2 obtained by approximating Eq. (2). Note that the approximate $C_{ei}^{(lk,0)}$ is nothing but the Lorentz collision operator, as expected. The collision operator for like species, C_{ee} and C_{ii} , can be easily obtained by setting $\mu=1, \theta=1$, and hence $\chi=1$. The results can be used in studying the unified fluid/kinetic description of plasma microinstabilities by treating the collision operators more precisely and rigorously as well as including higher order collisional terms.²⁶

D. Gyrophase average

Before closing this section, we discuss the gyrophase average of collision operator. A kinetic equation for a magnetized plasma can be simplified by suppressing the details of the short-length and fast-time scales through the gyrophase average. In the derivation of the drift-kinetic equation given in Ref. 18, it was assumed that

$$C(\langle f \rangle_\gamma) = \langle C(f) \rangle_\gamma \quad (51)$$

where $\langle \rangle_\gamma$ denotes the gyrophase average. Here we prove Eq. (51) for the linearized collision operator. Since there is no angle dependence in $C(f_a^{(0)}, f_b^{(0)})$ it trivially satisfies (51).

For the test-particle operator $C(f_a^{lk}, f_b^{(0)})$, the γ dependence appears only through f_a^{lk} and $L^2 f_a^{lk}$ in Eq. (43a). Since $L^2 = -\partial_\xi(1-\xi^2)\partial_\xi - (1-\xi^2)^{-1}\partial_\gamma^2$ and $\langle f_a^{lk} \rangle_\gamma = f_{a||}^{lk} P_l(\xi)$ with the speed-only dependent function $f_{a||}^{lk} = f_a^{(0)} L_k^{(l+1/2)} s_a^l P^l(\hat{\mathbf{b}}) \cdot \hat{\mathbf{M}}_a^{lk}$, we have

$$L^2 \langle f_a^{lk} \rangle_\gamma = l(l+1) \langle f_a^{lk} \rangle_\gamma \quad (52)$$

and hence

$$C(\langle f_a^{lk} \rangle_\gamma, f_b^{(0)}) = \langle C(f_a^{lk}, f_b^{(0)}) \rangle_\gamma. \quad (53)$$

For the field operator $C(f_a^{(0)}, f_b^{lk})$, there are no derivatives with respect to γ in Eq. (43b). In the expressions (9), (23) with (32), (35) with (34), and (44b), the γ -dependence of f_b^{lk} ,

G_b^{lk} , H_b^{lk} , and $C_{ab}^{(0,lk)}$ arises from the harmonic polynomial term $P^l(\mathbf{v})$ only. From the averaging property (20), $\langle P^l(\hat{\mathbf{v}}) \rangle_\gamma = P_l(\xi) P^l(\hat{\mathbf{b}})$, it is obvious that

$$C(f_a^{(0)}, \langle f_b^{lk} \rangle_\gamma) = \langle C(f_a^{(0)}, f_b^{lk}) \rangle_\gamma. \quad (54)$$

This completes the proof of (51) for the linearized collision operator.

For the nonlinear term, Eq. (51) no longer holds simply because the gyroaverage of the product of two γ -dependent functions is not the same as the product of the two gyroaveraged functions. This nonlinear term is significant when the system is far from equilibrium, that is, f^{lk} is comparable to $f^{(0)}$.

III. MOMENTS OF THE COLLISION OPERATOR

We find fluid moment equations by multiplying Eq. (1) by a velocity polynomial and integrating. A collisional moment is the velocity integral of the product of a velocity polynomial and the collision operator. In the collisional transport theory, the collisional moments of low orders have been calculated in the case of like-species and the small mass ratio approximation.^{1,2} In this section we find the collisional moments of arbitrary orders for general masses and temperatures, and hence write the most general fluid equations. Furthermore we can easily check the conservation properties of the operator which are tied to the underlying Coulomb collision process.

A. Calculation of exact collisional moments

In order to calculate the collisional moments, multiplying (44) by (11) and integrating by using the orthogonality relation (18), we can write

$$\int d\mathbf{v} P_a^{lp} C_{ab}^{(lk,0)} = A_{ab}^{lpk} \hat{\mathbf{M}}_a^{lk}, \quad (55a)$$

$$\int d\mathbf{v} P_a^{lp} C_{ab}^{(0,lk)} = B_{ab}^{lpk} \hat{\mathbf{M}}_b^{lk}, \quad (55b)$$

where we defined the coefficients

$$A_{ab}^{lpk} = \int d\mathbf{v} s_a^l L_p^{(l+1/2)}(s_a^2) f_a^{(0)} \nu_{ab}^{(lk,0)}, \quad (56a)$$

$$B_{ab}^{lpk} = \int d\mathbf{v} s_a^l L_p^{(l+1/2)}(s_a^2) f_a^{(0)} \nu_{ab}^{(0,lk)}. \quad (56b)$$

Furthermore from the expansions of Sonine polynomials (12) and (45), we can express the coefficients with Sonine polynomial indices p and k :

$$A_{ab}^{lpk} = \sum_{q=0}^p c_{pq}^l \sum_{m=0}^k c_{km}^l A_{ab}^{lqm}, \quad (57a)$$

$$B_{ab}^{lpk} = \sum_{q=0}^p c_{pq}^l \sum_{m=0}^k c_{km}^l B_{ab}^{lqm} \quad (57b)$$

in terms of the coefficients with monomial indices q and m :

$$A_{*ab}^{lqm} = \int d\mathbf{v} s_a^{l+2q} f_a^{(0)} v_{*ab}^{(lm,0)}, \quad (58a)$$

$$B_{*ab}^{lqm} = \int d\mathbf{v} s_a^{l+2q} f_a^{(0)} v_{*ab}^{(0,lm)}. \quad (58b)$$

Finally, a collisional moment is a linear combination of moments of the distribution function, $\hat{M}^{lk}$:

$$\int d\mathbf{v} \mathbf{P}_a^{lp} C(f_a, f_b) = \sum_k \frac{1}{\lambda_k} (A_{ab}^{lpk} \hat{M}_a^{lk} + B_{ab}^{lpk} \hat{M}_b^{lk}). \quad (59)$$

In the calculation of (58) using (46) and (48), we need to perform the following integrations:

$$e_{ab}^k = \int_0^\infty ds_b E_a' s_b^{2k} e^{-s_b^2}, \quad (60)$$

$$E_{ab}^k = \int_0^\infty ds_b E_a s_b^{2k+1} e^{-s_b^2}. \quad (61)$$

The integral (60) is nothing but a Gaussian function integral and gives

$$e_{ab}^k = \frac{(k-1/2)!}{(-1/2)!} X_{ab}^{k+1/2}, \quad (62)$$

where $X_{ab} = 1/(1+\chi^2)$. For Eq. (61), the integration by parts gives the following recurrence relation:

$$E_{ab}^k = k E_{ab}^{k-1} + \frac{\chi}{2} e_{ab}^k, \quad (63)$$

from which we can easily derive

$$E_{ab}^k = \frac{\chi}{2} \sum_{j=0}^k \frac{k!}{j!} e_{ab}^j. \quad (64)$$

Now we can write the coefficients for the moments of the collision operator

$$A_{*ab}^{lqm} = \sum_{n=0}^2 (a_{Enab}^{lm} \alpha_{Enab}^{lqm} + a_{enab}^{lm} \alpha_{enab}^{lqm}), \quad (65a)$$

$$B_{*ab}^{lqm} = \sum_{n=0}^1 (b_{enab}^{lm} \beta_{enab}^{lqm} + b_{+nab}^{lm} \beta_{+nab}^{lqm} + b_{-nab}^{lm} \beta_{-nab}^{lqm}), \quad (65b)$$

where

$$\alpha_{Enab}^{lqm} = \frac{4n_a v_{ab}}{\pi^{1/2}} E_{ba}^{l+q+m-n}, \quad (66a)$$

$$\alpha_{enab}^{lqm} = \frac{4n_a v_{ab}}{\pi^{1/2}} \frac{1}{\chi} e_{ba}^{l+q+m-n+1}, \quad (66b)$$

and

$$\beta_{enab}^{lqm} = \frac{4n_a v_{ab}}{\pi^{1/2}} \chi^{l+2(q+2)} e_{ab}^{l+q+m+n+1}, \quad (67a)$$

$$\beta_{+nab}^{lqm} = \frac{4n_a v_{ab}}{\pi^{1/2}} \chi^{l+2(1-n)} \times \frac{1}{2} \sum_{j=0}^{q+n} \frac{(q+n)!}{j!} \chi^{2j} e_{ab}^{l+m+j+1}, \quad (67b)$$

$$\beta_{-nab}^{lqm} = \frac{4n_a v_{ab}}{\pi^{1/2}} \chi^{l+2(q+2)} \frac{1}{2} \sum_{j=0}^m \frac{m!}{j!} e_{ab}^{l+q+n+j+1}. \quad (67c)$$

Here we have provided simple formulas of collisional moments for any arbitrary moments. Note that a collisional moment is a linear combination of the moments of the distribution function $\hat{M}^{lk}$ [see Eq. (55)] and the formulas (57) and (65) with (66) and (67) provide the collisional coefficients for the moments. Some collisional coefficients A_{ab}^{lpk} and B_{ab}^{lpk} for low orders $l=0, 1, 2$, $k=0, 1, 2$ are given in Appendix B.

For like species, our exact calculation includes the same results of low-order ($l=1, 2$) moments as Refs. 1, 3, and 4. However for unlike species our results disagree with Refs. 3 and 4. Since the linearized Coulomb collision operator is approximately self-adjoint for general mass and temperature ratios,^{5,6} the results of Refs. 3 and 4 obtained from the self-adjoint operator cannot be exact. Note that the generating formula, Eq. (A2) of Ref. 3 is missing the term proportional to $(1-T_a/T_b)$, in comparison to Eq. (A.4) of Ref. 1. We have independently obtained the $l=1$ collisional moments by calculating coefficients of $\xi^p \eta^k$ ($p, k=0, 1, 2$) from Eq. (A.4) of Ref. 1 and verified that they well agree with our results, Eqs. (B5)–(B7) and (B14)–(B16), in Appendix B.

As we discussed in the previous section, due to the explicit appearance of the mass ratio μ in the equations, the above formulas also easily provide for the small mass ratio approximation. The results for low order moments in Appendix C are the same as the original works of Braginskii.² It is also easy to write the collisional coefficients for like species as we will see in Sec. IV B.

B. Conservation laws

For a conserved quantity $\psi(\mathbf{v})$, the conservation laws can be expressed as

$$\int d\mathbf{v} \psi_a(\mathbf{v}) C(f_a, f_b) + \int d\mathbf{v} \psi_b(\mathbf{v}) C(f_b, f_a) = 0. \quad (68)$$

It is obvious from the symmetric form of the Landau-Boltzmann collision operator (2) that the conservation laws hold for the particle number $\psi=1$, the momentum $m\mathbf{v}$, and the energy $\frac{1}{2}m\mathbf{v}^2$. Since the distribution functions f_a and f_b are arbitrary in Eq. (68), it must hold, when $f_a = f_a^{lm} = f_a^{(0)} \mathbf{P}^l(\mathbf{s}_a) s_a^{2m} \cdot \hat{\mathbf{M}}_a^{lk}$ and $f_b = f_b^{(0)}$, for the linear collision operators:

$$\int d\mathbf{v} \psi_a C(f_a^{lm}, f_b^{(0)}) + \int d\mathbf{v} \psi_b C(f_b^{(0)}, f_a^{lm}) = 0. \quad (69)$$

However, it is not clear from investigating the form of (46) and (48) that the collision operator (43) should satisfy Eq. (69). In this section we will prove the conservation laws explicitly.

1. Particle conservation

For $\psi_a = 1$, the particle conservation law (69) can be written as

$$A_{*ab}^{00m} \hat{M}_a^{0k} + B_{*ba}^{00m} \hat{M}_a^{0k} = 0 \quad (70)$$

for the linear collision operator. In fact, the particle conservation satisfies a stronger form $\int d\mathbf{v} C(f_a, f_b) = 0$ which can be written as

$$A_{*ab}^{00m} = 0 = B_{*ab}^{00m}. \quad (71)$$

From Eq. (65a), it is easy to see that $A_{*ab}^{000} = 0, A_{*ab}^{001} = 0$. For $m \geq 2$, it follows from (63) that

$$A_{*ab}^{00m} = a_E E_{ba}^{m-2} + a_e e_{ba}^{m-1} \quad (72)$$

with vanishing coefficients a_E and a_e . In Eq. (65b), we can show that

$$\sum_{n=0}^1 (b_{enab}^{0m} \beta_{enab}^{00m} + b_{+nab}^{0m} \beta_{+nab}^{00m}) = \frac{4n_a \nu_{ab}}{\pi^{1/2}} \frac{4\theta}{3\mu} e_{ab}^{m+2} \quad (73)$$

and the remaining part of (65b) can be written as

$$\sum_{n=0}^1 b_{-nab}^{0m} \beta_{-nab}^{00m} = -\frac{4n_a \nu_{ab}}{\pi^{1/2}} \frac{4\theta}{3\mu} e_{ab}^{m+2} \quad (74)$$

from the following identity:

$$\sum_{j=0}^m \frac{1}{j!} \left(-e_{ab}^{j+1} + \frac{2\theta}{3\mu} e_{ab}^{j+2} \right) = -\frac{2}{3} \frac{e_{ab}^{m+2}}{m!}. \quad (75)$$

Therefore, we have proved Eq. (71). One can also verify the particle conservation law from Eqs. (A3c), (B2), and (B11) in the Appendices.

2. Energy conservation

For $\psi_a = \frac{1}{2} m_a v^2$, the energy conservation law can be written as

$$T_a A_{*ab}^{01m} \hat{M}_a^{0k} + T_b B_{*ba}^{01m} \hat{M}_a^{0k} = 0. \quad (76)$$

From the expressions (65) with $l=0$ and $q=1$, after some algebra we can write

$$A_{*ab}^{01m} = \frac{4n_a \nu_{ab}}{\pi^{1/2}} \left\{ -\frac{2m}{\mu} E_{ba}^{m-1} + \left[1 - \frac{1}{\theta} + 2m + \frac{2m}{\mu} \right] \times \frac{1}{\chi} X_{ba} e_{ba}^m \right\}, \quad (77a)$$

$$B_{*ab}^{01m} = \frac{4n_a \nu_{ab}}{\pi^{1/2}} \left\{ \frac{2m}{\chi} E_{ab}^{m-1} - \left[2m + (1 - \theta + 2m) \frac{1}{\mu} \right] \times X_{ab} e_{ab}^m \right\}. \quad (77b)$$

Noting that $\nu_{ba} = \frac{1}{\mu\theta\chi} \nu_{ab}$ and

$$B_{*ba}^{01m} = \frac{4n_a \nu_{ba}}{\pi^{1/2}} \left\{ 2m \chi E_{ba}^{m-1} - \left[2m + \left(1 - \frac{1}{\theta} + 2m \right) \mu \right] \times X_{ba} e_{ba}^m \right\}$$

one can easily prove (76). Before we proceed to prove the momentum conservation, performing the summation over m in

$$A_{ab}^{01k} = \sum_{m=0}^k c_{km}^0 c_{11}^0 A_{*ab}^{01m} = - \sum_{m=0}^k c_{km}^0 A_{*ab}^{01m}, \quad (78)$$

we can further simplify the coefficient for the collisional heating:

$$A_{ab}^{01k} = -\frac{4n_a \nu_{ab}}{\pi^{1/2}} \frac{(k-1/2)!}{(-1/2)! k!} \left[2k \left(1 + \frac{1}{\mu} \right) + 1 - \frac{1}{\theta} \right] \times X_{ba} X_{ab}^{k+1/2}. \quad (79)$$

This equation will be useful in the calculation of collisional heating. One can also verify the energy conservation law from Eqs. (A4c), (B3), and (B12) in the Appendices.

3. Momentum conservation

Similarly, for $\psi_a = m_a \mathbf{v}$ the momentum conservation means

$$m_a v_{Ta} A_{ab}^{10m} \hat{M}_a^{1k} + m_b v_{Tb} B_{ba}^{10m} \hat{M}_a^{1k} = 0. \quad (80)$$

From (65) with $l=1$ and $q=0$,

$$A_{*ab}^{10m} = \frac{4n_a \nu_{ab}}{\pi^{1/2}} \left\{ -m \left(1 + \frac{1}{\mu} \right) E_{ba}^{m-1} + \left[m \left(1 + \frac{1}{\mu} \right) - \frac{1}{2\theta} (1 + \mu) \right] \frac{1}{\chi} X_{ba} e_{ba}^m \right\}, \quad (81a)$$

$$B_{*ab}^{10m} = \frac{4n_a \nu_{ab}}{\pi^{1/2}} \left\{ \frac{m}{\theta} (1 + \mu) E_{ab}^{m-1} - \left[\frac{m}{\theta} (1 + \mu) - \frac{1}{2} \left(1 + \frac{1}{\mu} \right) \right] \chi X_{ab} e_{ab}^m \right\}. \quad (81b)$$

It is easy to prove (80) from

$$B_{*ba}^{10m} = \frac{4n_a \nu_{ba}}{\pi^{1/2}} \left\{ m \theta \left(1 + \frac{1}{\mu} \right) E_{ba}^{m-1} - \left[m \theta \left(1 + \frac{1}{\mu} \right) - \frac{1}{2} (1 + \mu) \right] \frac{1}{\chi} X_{ba} e_{ba}^m \right\}. \quad (82)$$

The simple form of the frictional force coefficient is

$$A_{ab}^{10k} = -\frac{4n_a \nu_{ab}}{\pi^{1/2}} \frac{(k+1/2)!}{k! (-1/2)!} \left(1 + \frac{1}{\mu} \right) X_{ab}^{k+3/2}. \quad (83)$$

This equation together with the momentum conservation law will be useful in the calculation of collisional friction force and hence in the transport theory. One can also verify the momentum conservation law from Eqs. (A5c), (B5), and (B14) in the Appendices.

IV. MOMENTS OF THE KINETIC EQUATION

A. General moment equation

In this section we consider the moments of the kinetic equation. The differential equations for the fluid moments are obtained by substituting the expansion (8) into the Boltzmann equation (1) with the Coulomb collision operator, multiplying by a proper velocity polynomial (11) and integrating over velocity.

Here is the moment equation in a matrix form:

$$\begin{pmatrix} D^0 & D^{0+} & 0 & 0 & \cdots \\ D^{1-} & D^1 & D^{1+} & 0 & \cdots \\ 0 & D^{2-} & D^2 & D^{2+} & \cdots \\ 0 & 0 & D^{3-} & D^3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} M_a^0 \\ M_a^1 \\ M_a^2 \\ M_a^3 \\ \vdots \end{pmatrix} = \sum_b \begin{pmatrix} \bar{A}_{ab}^0 M_a^0 \\ \bar{A}_{ab}^1 M_a^1 \\ \bar{A}_{ab}^2 M_a^2 \\ \bar{A}_{ab}^3 M_a^3 \\ \vdots \end{pmatrix} + \sum_b \begin{pmatrix} \bar{B}_{ab}^0 M_b^0 \\ \bar{B}_{ab}^1 M_b^1 \\ \bar{B}_{ab}^2 M_b^2 \\ \bar{B}_{ab}^3 M_b^3 \\ \vdots \end{pmatrix}, \quad (84)$$

where M^l is a column vector of moments $(M^{l0}, M^{l1}, M^{l2}, \dots)$, $\bar{A}_{ab}^{lpk} = (n_a \lambda_k^l)^{-1} v_{Ta}^{2(p-k)} A_{ab}^{lpk}$ and $\bar{B}_{ab}^{lpk} = (n_a \lambda_k^l)^{-1} v_{Ta}^{l+2p} v_{Tb}^{-l-2k} B_{ab}^{lpk}$. The submatrices D^l and $D^{l\pm}$ are

$$D^l = \partial_t + (\partial_t \ln n_a) + \Xi^l (\partial_t \ln T) + l \Omega_a \hat{b} \times \quad (85a)$$

$$D^{l\pm} = \Psi^{l\pm} \nabla + \Psi^{l\pm} \nabla \ln n_a + \Phi^{l\pm} (\nabla \ln T) + \frac{e_a}{m_a} \Theta^{l\pm} E \quad (85b)$$

where Ξ^l , $\Phi^{l\pm}$, $\Psi^{l\pm}$, and $\Theta^{l\pm}$ are submatrices acting on a vector M^l . The matrix elements for the p th row and k th column are given by

$$\Xi_{pk}^l = -v_T^2 (l+p+1/2) \delta_{p-1,k}, \quad (86a)$$

$$\Phi_{pk}^{l-} = \frac{l(l+p+1/2)}{2l+1} [v_T^2 (p+1) \delta_{p,k} - v_T^4 (l+p-1/2) \delta_{p,k+1}], \quad (86b)$$

$$\Phi_{pk}^{l+} = -v_T^2 (l+p+3/2) \delta_{p,k+1} + v_T^4 (l+p+1/2) \delta_{p,k+2}, \quad (86c)$$

$$\Psi_{pk}^{l-} = \frac{l}{2l+1} [v_T^2 (l+p+1/2) \delta_{p,k} - (p+1) \delta_{p,k-1}], \quad (86d)$$

$$\Psi_{pk}^{l+} = \delta_{p,k} - v_T^2 \delta_{p-1,k}, \quad (86e) \quad \text{and}$$

$$\Theta_{pk}^{l-} = -\frac{l(l+p+1/2)}{l+1/2} \delta_{p,k}, \quad (86f)$$

$$\Theta_{pk}^{l+} = 2 \delta_{p-1,k}. \quad (86g)$$

B. Single component ionized gas

To study the properties of the collision matrix we consider here a single component ionized gas which has a simple form of the collision matrix. The study of a single component gas does not only show important properties of the collisional matrix but also provides for methods to get closure relations depending on the ordering schemes. Furthermore the moment equation for ions in a pure plasma is effectively the same as the equation for the single component ionized gas since the ion-electron collision operator is negligible compared to the ion-ion collision operator in the small mass ratio approximation (see Appendix C).

Defining the like-species collision matrix $C^l = A_{aa}^l + B_{aa}^l$, the right-hand side of Eq. (84) is $\bar{C}^l M^l$ with $\bar{C}^{lpk} = (n_a \lambda_k^l)^{-1} v_T^{2(p-k)} C^{lpk}$, where we suppressed species subscripts. Here are the collision matrices for $l=0, 1, 2$:

$$C^0 = \frac{-3n_a}{2\sqrt{2}\tau_{aa}} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{4} & \frac{15}{32} \\ 0 & 0 & \frac{3}{4} & \frac{31}{16} & \frac{201}{128} \\ 0 & 0 & \frac{15}{32} & \frac{201}{128} & \frac{2929}{1024} \end{pmatrix}, \quad (87)$$

$$C^1 = \frac{-3n_a}{\sqrt{2}\tau_{aa}} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & \frac{3}{4} & \frac{15}{32} & \frac{35}{128} \\ 0 & \frac{3}{4} & \frac{45}{16} & \frac{309}{128} & \frac{885}{512} \\ 0 & \frac{15}{32} & \frac{309}{128} & \frac{5657}{1024} & \frac{20349}{4096} \\ 0 & \frac{35}{128} & \frac{885}{512} & \frac{20349}{4096} & \frac{149749}{16384} \end{pmatrix}, \quad (88)$$

$$C^2 = \frac{-9n_a}{2\sqrt{2}\tau_{aa}} \begin{pmatrix} 1 & \frac{3}{4} & \frac{15}{32} & \frac{35}{128} & \frac{315}{2048} \\ \frac{3}{4} & \frac{205}{48} & \frac{489}{128} & \frac{1435}{512} & \frac{46235}{24576} \\ \frac{15}{32} & \frac{489}{128} & \frac{11889}{1024} & \frac{45129}{4096} & \frac{573045}{65536} \\ \frac{35}{128} & \frac{1435}{512} & \frac{45129}{4096} & \frac{1220437}{49152} & \frac{6323017}{262144} \\ \frac{315}{2048} & \frac{46235}{24576} & \frac{573045}{65536} & \frac{6323017}{262144} & \frac{575512075}{12582912} \end{pmatrix}, \quad (89)$$

where $\tau_{ab} = 3\pi^{1/2}/4\nu_{ab}$.

The above collision matrices apparently show an important aspect of Boltzmann's H -theorem. To simply discuss this we treat temperature and density perturbatively by writing $T = T_0 + \delta T$ ($\delta T \ll T_0$) and $n = n_0 + \delta n$ ($\delta n \ll n_0$) (hereafter a species index is suppressed). In this perturbation theory it is convenient to use constant density n_0 , constant temperature T_0 , and constant thermal speed $v_0 = (2T_0/m)^{1/2}$ for defining a normalized velocity $\mathbf{s}_0 = \mathbf{v}/v_0$, a background Maxwellian distribution in Eq. (9) $f^{(0)} = \frac{n_0}{\pi^{3/2}v_0^3} e^{-s_0^2}$, moments $n_0 M_0^{lk}(t, \mathbf{x}) = \int d\mathbf{v} \mathbf{P}^l(\mathbf{v}) v_0^{2k} L_k^{(l+1/2)}(s^2) f(t, \mathbf{x}, \mathbf{v})$. To the lowest order, the matrices in the moment equation (84) have neither spatial nor temporal derivatives of temperature and density and, instead, these derivatives appear through derivatives of moments $M_0^{00} = 1 + \frac{\delta n}{n_0}$ and $M_0^{01} \approx -\frac{3}{m} \delta T - V^2$ in the perturbation theory. Furthermore, assuming the spatial uniformity reduces Eq. (84) to $\partial_t M_0^l = \bar{C}^l M_0^l$. [One can derive the same equation directly from Eq. (84) absorbing temperature and density gradient terms into M^{lk} appropriately and hence redefining M_0^{lk} .] Obviously the solutions show that the density M_0^{00} , the temperature M_0^{01} , and the fluid velocity $M_0^{10} = \mathbf{V}$ corresponding to three zero eigenvalues do not change in time and all the other moments exponentially decay due to their negative eigenvalues. Consequently, a system with any arbitrary distribution approaches the Maxwellian distribution and then stays in equilibrium.

C. Collisional heat flux equation

When the ratio of the mean free path to the gradient scale is very small ($\epsilon \ll 1$), the expansion of the moments

$$M^l = M_{(0)}^l + \epsilon M_{(1)}^l + \epsilon^2 M_{(2)}^l + \dots \quad (90)$$

provides for collisional closures. This expansion is merely a manifestation of the Chapman-Enskog-type solution $f = f^{(0)} + \epsilon f^{(1)} + \epsilon^2 f^{(2)} + \dots$ in the moment expansion (8). For simplicity, we consider a stationary system with the temperature gradient term only in Eq. (85):

$$D^l = 0, \quad D^{\pm} = \Psi^{\pm} \nabla + \Phi^{\pm} (\nabla \ln T). \quad (91)$$

Keeping $D^{\pm} \sim \epsilon C^l$ in mind, the zeroth order equation

$$\bar{C}^l M_{(0)}^l = 0 \quad (92)$$

has nontrivial solutions for $M_{(0)}^{00}$, $M_{(0)}^{01}$, and $M_{(0)}^{10}$ only. Assuming the Maxwellian distribution $f^{(0)}$ as the zeroth order solution, we have $M_{(0)}^{lk} = 0$ except for $M_{(0)}^{00} = 1$. Subsequently, the nontrivial first-order solutions are calculated from

$$\bar{C}^1 M_{(1)}^1 = D^1 - M_{(0)}^0, \quad (93)$$

the second-order solutions are from

$$\bar{C}^0 M_{(2)}^0 = D^{0+} \cdot M_{(1)}^1, \quad (94)$$

$$\bar{C}^2 M_{(2)}^2 = D^{2-} M_{(1)}^1, \quad (95)$$

and the third-order solutions are from

$$\bar{C}^1 M_{(3)}^1 = D^{1-} M_{(2)}^0 + D^{1+} \cdot M_{(2)}^2, \quad (96)$$

$$\bar{C}^3 M_{(3)}^3 = D^{3-} M_{(2)}^2. \quad (97)$$

The remarkable feature of the ordering (90) is the convergence of the results with increasing order of moment expansions $L_k^{(3/2)}$ as discussed below.

From the 8×8 ($k=7$) matrix calculation, we show the final result for the ion heat flux $\mathbf{q}_i \equiv -\frac{1}{2} m_i n_i M_i^{11}$,

$$\begin{aligned} \mathbf{q}_i = & -\kappa_{i\parallel} \nabla T_i - \kappa_{i1} (\nabla T_i)^2 \nabla T_i - \kappa_{i2} T_i \nabla^2 T_i \nabla T_i \\ & - \kappa_{i2} T_i \nabla \nabla T_i \cdot \nabla T_i - \kappa_{i3} T_i^2 \nabla \nabla^2 T_i, \end{aligned} \quad (98)$$

where

$$\kappa_{i\parallel} = 3.950 \frac{n_i \tau_i T_i}{m_i}, \quad (99a)$$

$$\kappa_{i1} = 4.633 \times 10^3 \frac{n_i \tau_i^3}{m_i^2}, \quad (99b)$$

$$\kappa_{i2} = 1.215 \times 10^3 \frac{n_i \tau_i^3}{m_i^2}, \quad (99c)$$

$$\kappa_{i2} = 1.304 \times 10^3 \frac{n_i \tau_i^3}{m_i^2}, \quad (99d)$$

$$\kappa_{i3} = 1.277 \times 10^2 \frac{n_i \tau_i^3}{m_i^2} \quad (99e)$$

with $\tau_i = \tau_{ii}/\sqrt{2}$. Note that the order of the Braginskii equation is ϵ^1 and the next higher order is ϵ^3 . For increasing moment expansion order $k=2$ to $k=3$, while the Braginskii coefficient $\kappa_{i||} = 3.906 n_i \tau_i T_i / m_i$, Eq. (4.40) in Ref. 2, is improved to $\kappa_{i||}(k=3) = 3.950 \frac{n_i \tau_i T_i}{m_i}$ by about 1%, and the ϵ^3 -order coefficient $\kappa_{i\perp}(k=2) = 2.052 \times 10^3 \frac{n_i \tau_i^3}{m_i^2}$ is improved to $\kappa_{i\perp}(k=3) = 4.135 \times 10^3 \frac{n_i \tau_i^3}{m_i^2}$ by a factor of 2. Thus the higher-order moment calculation is needed to have converging ϵ^3 -order transport coefficients. We have verified that $k=5$ calculation improves the above $k=4$ calculation by less than 1% and the $k=6$ calculation improves the $k=5$ calculation by about 0.2%. The $k=8$ calculation does not change the above values in four significant figures.

Finally we discuss the physical meaning of the ϵ^3 terms. While the ϵ^1 term involves the slope in the temperature along the direction of ∇T , the ϵ^3 terms involve additionally the curvatures along and perpendicular to the ∇T . In the ϵ^3 -order calculation of transport equations, we also clearly see the limitation of Braginskii equations. Due to the large numerical values of the ϵ^3 -order coefficients, when $\epsilon \sim 0.01$ the ϵ^3 -order correction is about 10% of the ϵ^1 -order original Braginskii term and when $\epsilon \sim 0.03$ the correction is comparable to the original. Note that in the presence of the magnetic field the parallel component equations remove the magnetic field term in Eq. (84) and give the same relations as above. We expect that other parallel closures can be calculated similarly and have the same converging properties.

V. SUMMARY AND DISCUSSIONS

In this paper we have calculated the Rosenbluth potentials of the distribution function in the moment expansion using the properties of the irreducible harmonic polynomials and the Legendre polynomials. The analytical expression of the Rosenbluth potentials enable us to write the linearized Coulomb collision operator and the collisional moments in closed forms for any arbitrary moments specified by the orders of the irreducible harmonic polynomials and Sonine polynomials. Furthermore, the expressions are given in terms of arbitrary mass and temperature ratios for any two plasma species. We have also discussed the gyroaveraging properties of the collision operator related to the derivation of the drift-kinetic equation.

From the moments of collision operators we have shown explicitly the conservation laws for the particle number, momentum, and energy moments. Note that the conservation laws are satisfied by each moment term by term. We also have derived the simple expressions for the momentum and energy moments of the collision operator that are useful to write the collisional friction force and the collisional heating in the transport theory. The general moment equations in a matrix form clearly show the couplings between neighboring

moments due to the derivative operator D 's. However the collision operator couples all the moments together in the l th harmonic subspace, where l is the rank of the irreducible harmonic polynomials P^l . We have introduced a closure procedure that is analogous to the Chapman-Enskog procedure in the collisional plasma. In spite of the couplings through the collision operator, we show that the collisional transport equations are well-defined regardless of the truncation of the collision matrix.

In the short mean-free-path regime ($\epsilon \ll 1$), we can apply our procedure to find more general solutions of the Spitzer-Härm problem²⁷ for classical transport theory in the presence of multiple species of ions.³ We also expect that the collision operators and coefficients could be used for calculating the neoclassical transport of impurities in tokamak plasmas as in Ref. 4. The collision process between charged particles with comparable masses requires exact calculation of collision operators without small-mass approximation. The solutions can be obtained directly from Eq. (84) by taking proper components and following the same procedure of finding the heat flux.

In longer mean-free-path regime ($\epsilon \sim 1$) the converging procedure described above is not simple and the diffusive heat flux model will fail. Moreover, in a recent simulation of spheromak evolution and energy confinement,^{28,29} the coexistence of collisional and nearly collisionless regimes requires general closures that can capture both regimes of physics. In the previous works^{8,30-32} general parallel closures are derived using the pitch-angle scattering collision operator only, as opposed to the full collision operator presented here. The collision operator formula given here will facilitate the derivation of the general closures with the full collision operator since it is well known that the free streaming solution requires higher order moments in the expansion of the solution.

When the distribution function is not close to the Maxwellian distribution function one should not ignore the nonlinear collision operator $C^{(2)}$ in Eq. (42). Since we have the explicit form of Rosenbluth potentials in the moment expansion, we can also calculate the nonlinear collision operators simply by differentiations in Eq. (40). Further investigations will include the effect of nonlinear collision terms in the approach to equilibrium from an arbitrary nonequilibrium initial distribution.

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APPENDIX A: SPEED FUNCTIONS OF THE COLLISION OPERATOR

We define the speed functions ν^A and ν^B for the test-particle operator and the field operator by

$$\nu_{ab}^{(lk,0)} = \frac{\nu_{ab}}{\sqrt{\theta\mu}} \nu_{ab}^{Alk}, \quad (\text{A1a})$$

$$\nu_{ab}^{(0,lk)} = \frac{\nu_{ab}}{\sqrt{\theta\mu}} \nu_{ab}^{Blk}. \quad (\text{A1b})$$

For like species ($\theta=1$, $\mu=1$), we also define

$$\nu^{jk} = \nu_{aa}^{(lk,0)} + \nu_{aa}^{(0,lk)}, \quad (\text{A2})$$

where the species subscript will be suppressed.

Here are the speed functions for some low order moments:

$$\nu_{ab}^{A00} = 2(1-\theta) \left[-\frac{E_b}{s_b} + \left(1 + \frac{\mu}{\theta}\right) E'_b \right], \quad (\text{A3a})$$

$$\nu_{ab}^{B00} = \nu_{ab}^{A00}, \quad (\text{A3b})$$

$$\nu^{00} = 0, \quad (\text{A3c})$$

$$\begin{aligned} \nu_{ab}^{A01} = & [(-5+7\theta) + 2(1-\theta)s_a^2] \frac{E_b}{s_b} \\ & + \left[\left(5-5\mu-7\theta+3\frac{\mu}{\theta}\right) - 2(1-\theta) \left(1 + \frac{\mu}{\theta}\right) s_a^2 \right] E'_b, \end{aligned} \quad (\text{A4a})$$

$$\begin{aligned} \nu_{ab}^{B01} = & -2\theta \frac{E_b}{s_b} + \left[\left(-\mu+2\theta+3\frac{\mu}{\theta}\right) \right. \\ & \left. - 2(1-\theta) \left(1 + \frac{\mu}{\theta}\right) s_b^2 \right] E'_b, \end{aligned} \quad (\text{A4b})$$

$$\nu^{01} = 0, \quad (\text{A4c})$$

$$\begin{aligned} \nu_{ab}^{A10} = & [(1-\mu-2\theta)s_a^{-1} - 2(1-\theta)s_a] \frac{E_b}{s_b} \\ & + \left[(-1+\mu+2\theta)s_a^{-1} + 2(1-\theta) \left(1 + \frac{\mu}{\theta}\right) s_a \right] E'_b, \end{aligned} \quad (\text{A5a})$$

$$\begin{aligned} \nu_{ab}^{B10} = & (-2+\mu+3\theta)s_b^{-1} \frac{E_b}{s_b} \\ & + \left[(2-\mu-3\theta)s_b^{-1} + 2(1-\theta) \left(1 + \frac{\mu}{\theta}\right) s_b \right] E'_b, \end{aligned} \quad (\text{A5b})$$

$$\nu^{10} = 0, \quad (\text{A5c})$$

$$\begin{aligned} \nu_{ab}^{A11} = & \left[\left(\frac{5}{2} - \frac{5}{2}\mu - 7\theta \right) s_a^{-1} + (-8+\mu+11\theta)s_a \right. \\ & + 2(1-\theta)s_a^3 \left. \right] \frac{E_b}{s_b} + \left[\left(-\frac{5}{2} + \frac{5}{2}\mu + 7\theta \right) s_a^{-1} \right. \\ & + \left(8-8\mu-11\theta+5\frac{\mu}{\theta} \right) s_a - 2(1-\theta) \\ & \left. \times \left(1 + \frac{\mu}{\theta} \right) s_a^3 \right] E'_b, \end{aligned} \quad (\text{A6a})$$

$$\begin{aligned} \nu_{ab}^{B11} = & -3\theta s_b^{-1} \frac{E_b}{s_b} + \left[3\theta s_b^{-1} + \left(1-2\mu+2\theta+5\frac{\mu}{\theta} \right) s_b \right. \\ & \left. - 2(1-\theta) \left(1 + \frac{\mu}{\theta} \right) s_b^3 \right] E'_b, \end{aligned} \quad (\text{A6b})$$

$$\nu^{11} = \left(-\frac{10}{s} + 4s \right) \frac{E}{s} + \frac{10}{s} E', \quad (\text{A6c})$$

$$\begin{aligned} \nu_{ab}^{A20} = & \left[\frac{3}{2}\theta s_a^{-2} + (2-3\mu-4\theta) - 2(1-\theta)s_a^2 \right] \frac{E_b}{s_b} \\ & + \left[-\frac{3}{2}\theta s_a^{-2} + (-2+2\mu+4\theta) + 2(1-\theta) \right. \\ & \left. \times \left(1 + \frac{\mu}{\theta} \right) s_a^2 \right] E'_b, \end{aligned} \quad (\text{A7a})$$

$$\begin{aligned} \nu_{ab}^{B20} = & \left[\left(-\frac{9}{2} + 3\mu + 9\theta \right) s_b^{-2} - \theta \right] \frac{E_b}{s_b} \\ & + \left[\left(\frac{9}{2} - 3\mu - 9\theta \right) s_b^{-2} + (3-2\mu-5\theta) + 2(1-\theta) \right. \\ & \left. \times \left(1 + \frac{\mu}{\theta} \right) s_b^2 \right] E'_b, \end{aligned} \quad (\text{A7b})$$

$$\nu^{20} = \left(\frac{9}{s^2} - 6 \right) \frac{E}{s} - \frac{9}{s^2} E', \quad (\text{A7c})$$

$$\begin{aligned} \nu_{ab}^{A21} = & \left[\frac{21}{4}\theta s_a^{-2} + \left(7 - \frac{21}{2}\mu - \frac{39}{2}\theta \right) \right. \\ & + (-11+3\mu+15\theta)s_a^2 + 2(1-\theta)s_a^4 \left. \right] \frac{E_b}{s_b} \\ & + \left[-\frac{21}{4}\theta s_a^{-2} + \left(-7+7\mu+\frac{39}{2}\theta \right) + (11-11\mu) \right. \\ & \left. - 15\theta+7\frac{\mu}{\theta} \right] s_a^2 - 2(1-\theta) \left(1 + \frac{\mu}{\theta} \right) s_a^4 \left. \right] E'_b, \end{aligned} \quad (\text{A8a})$$

$$\begin{aligned} \nu_{ab}^{B21} = & -9\theta s_b^{-2} \frac{E_b}{s_b} + \left[9\theta s_b^{-2} + 6\theta + \left(2-3\mu+2\theta+7\frac{\mu}{\theta} \right) s_b^2 \right. \\ & \left. - 2(1-\theta) \left(1 + \frac{\mu}{\theta} \right) s_b^4 \right] E'_b, \end{aligned} \quad (\text{A8b})$$

$$\nu^{21} = \left(-\frac{15}{4s^2} - 23 + 7s^2 \right) \frac{E}{s} + \left(\frac{15}{4s^2} + \frac{51}{2} \right) E'. \quad (\text{A8c})$$

APPENDIX B: VALUES OF COLLISION COEFFICIENTS

In this appendix, we write the collision coefficients for low order moments. We can verify the particle conservation $A_{ab}^{00k} = 0 = B_{ab}^{00k}$, the energy conservation $A_{ab}^{01k} + \theta B_{ba}^{01k} = 0$, and the momentum conservation $A_{ab}^{10k} + \mu \chi B_{ba}^{10k} = 0$ for $k=0, 1, 2$.

Defining a_{pk}^l and b_{pk}^l by

$$A_{ab}^{lpk} = -\frac{4n_a \nu_{ab}}{\pi^{1/2}} X_{ab}^{l+p+k+1/2} \left(\frac{\theta}{\mu} \right) a_{pk}^l, \quad (\text{B1a})$$

$$B_{ab}^{lpk} = \frac{4n_a \nu_{ab}}{\pi^{1/2}} X_{ab}^{l+p+k+1/2} \left(\frac{\theta}{\mu} \right)^{\frac{1}{2}+k} b_{pk}^l. \quad (\text{B1b})$$

Here are some values of them:

$$a_{0k}^0 = 0, \quad (\text{B2})$$

$$a_{10}^0 = 1 - \frac{1}{\theta}, \quad (\text{B3a})$$

$$a_{11}^0 = \frac{3}{2} - \frac{1}{2\theta} + \frac{1}{\mu}, \quad (\text{B3b})$$

$$a_{12}^0 = \frac{15}{8} - \frac{3}{8\theta} + \frac{3}{2\mu}, \quad (\text{B3c})$$

$$a_{20}^0 = \frac{3}{2} - \frac{3}{2\theta}, \quad (\text{B4a})$$

$$a_{21}^0 = \frac{13}{4} - \frac{7}{4\theta} - \frac{5\theta}{2\mu^2} + \frac{5\theta^2}{2\mu^2} - \frac{1}{2\mu} + \frac{2\theta}{\mu}, \quad (\text{B4b})$$

$$a_{22}^0 = \frac{69}{16} - \frac{17}{16\theta} + \frac{5\theta^2}{2\mu^3} - \frac{3\theta}{4\mu^2} + \frac{21\theta^2}{4\mu^2} + \frac{9}{4\mu} + \frac{3\theta}{\mu}, \quad (\text{B4c})$$

$$a_{00}^1 = \frac{1}{2\theta} + \frac{\mu}{2\theta}, \quad (\text{B5a})$$

$$a_{01}^1 = \frac{3}{4\theta} + \frac{3\mu}{4\theta}, \quad (\text{B5b})$$

$$a_{02}^1 = \frac{15}{16\theta} + \frac{15\mu}{16\theta}, \quad (\text{B5c})$$

$$a_{10}^1 = 1 - \frac{1}{4\theta} - \frac{5}{2\mu} + \frac{5\theta}{2\mu} + \frac{3\mu}{4\theta}, \quad (\text{B6a})$$

$$a_{11}^1 = \frac{5}{2} + \frac{9}{8\theta} + \frac{15\theta}{4\mu^2} - \frac{3}{4\mu} + \frac{13\theta}{2\mu} + \frac{13\mu}{8\theta}, \quad (\text{B6b})$$

$$a_{12}^1 = \frac{27}{8} + \frac{57}{32\theta} + \frac{63\theta}{8\mu^2} - \frac{3}{16\mu} + \frac{177\theta}{16\mu} + \frac{69\mu}{32\theta}, \quad (\text{B6c})$$

$$a_{20}^1 = \frac{3}{2} - \frac{9}{16\theta} - \frac{21}{4\mu} + \frac{21\theta}{4\mu} + \frac{15\mu}{16\theta}, \quad (\text{B7a})$$

$$a_{21}^1 = \frac{23}{4} - \frac{19}{32\theta} - \frac{35\theta^2}{4\mu^3} + \frac{35\theta^3}{4\mu^3} - \frac{21\theta}{8\mu^2} + \frac{21\theta^2}{2\mu^2} - \frac{87}{8\mu} + \frac{87\theta}{4\mu} + \frac{69\mu}{32\theta}, \quad (\text{B7b})$$

$$a_{22}^1 = \frac{161}{16} + \frac{233}{128\theta} + \frac{175\theta^3}{16\mu^4} - \frac{7\theta^2}{8\mu^3} + \frac{413\theta^3}{16\mu^3} + \frac{303\theta}{16\mu^2} + \frac{95\theta^2}{4\mu^2} - \frac{139}{32\mu} + \frac{1329\theta}{32\mu} + \frac{433\mu}{128\theta}, \quad (\text{B7c})$$

$$a_{00}^2 = 3 + \frac{1}{\theta} + \frac{5}{2\mu} + \frac{3\mu}{2\theta}, \quad (\text{B8a})$$

$$a_{01}^2 = 6 + \frac{3}{2\theta} + \frac{21}{4\mu} + \frac{9\mu}{4\theta}, \quad (\text{B8b})$$

$$a_{02}^2 = \frac{75}{8} + \frac{15}{8\theta} + \frac{135}{16\mu} + \frac{45\mu}{16\theta}, \quad (\text{B8c})$$

$$a_{10}^2 = 8 - \frac{1}{2\theta} - \frac{35\theta}{4\mu^2} + \frac{35\theta^2}{4\mu^2} - \frac{7}{4\mu} + \frac{7\theta}{\mu} + \frac{9\mu}{4\theta}, \quad (\text{B9a})$$

$$a_{11}^2 = \frac{105}{4} + \frac{13}{4\theta} + \frac{35\theta^2}{2\mu^3} + \frac{21\theta}{8\mu^2} + \frac{273\theta^2}{8\mu^2} + \frac{117}{8\mu} + \frac{111\theta}{4\mu} + \frac{51\mu}{8\theta}, \quad (\text{B9b})$$

$$a_{12}^2 = \frac{333}{8} + \frac{81}{16\theta} + \frac{189\theta^2}{4\mu^3} + \frac{351\theta}{32\mu^2} + \frac{2349\theta^2}{32\mu^2} + \frac{891}{32\mu} + \frac{189\theta}{4\mu} + \frac{279\mu}{32\theta}, \quad (\text{B9c})$$

$$a_{20}^2 = \frac{99}{8} - \frac{9}{8\theta} - \frac{189\theta}{8\mu^2} + \frac{189\theta^2}{8\mu^2} - \frac{81}{16\mu} + \frac{27\theta}{2\mu} + \frac{45\mu}{16\theta}, \quad (\text{B10a})$$

$$a_{21}^2 = \frac{387}{8} - \frac{27}{16\theta} - \frac{315\theta^3}{8\mu^4} + \frac{315\theta^4}{8\mu^4} - \frac{63\theta^2}{4\mu^3} + \frac{63\theta^3}{\mu^3} - \frac{1161\theta}{16\mu^2} + \frac{2511\theta^2}{16\mu^2} - \frac{297}{32\mu} + \frac{675\theta}{8\mu} + \frac{279\mu}{32\theta}, \quad (\text{B10b})$$

$$a_{22}^2 = \frac{6543}{64} + \frac{441}{64\theta} + \frac{945\theta^4}{16\mu^5} + \frac{189\theta^3}{16\mu^4} + \frac{2457\theta^4}{16\mu^4} + \frac{1323\theta^2}{8\mu^3} + \frac{3213\theta^3}{16\mu^3} + \frac{441\theta}{64\mu^2} + \frac{26163\theta^2}{64\mu^2} + \frac{5733}{128\mu} + \frac{1629\theta}{8\mu} + \frac{2115\mu}{128\theta}, \quad (\text{B10c})$$

$$b_{0k}^0 = 0, \quad (\text{B11})$$

$$b_{10}^0 = \frac{1}{\mu} - \frac{\theta}{\mu}, \quad (\text{B12a})$$

$$b_{11}^0 = 1 + \frac{3}{2\mu} - \frac{\theta}{2\mu}, \quad (\text{B12b})$$

$$b_{12}^0 = \frac{3}{2} + \frac{15}{8\mu} - \frac{3\theta}{8\mu}, \quad (\text{B12c})$$

$$b_{20}^0 = \frac{3}{2\mu} - \frac{3\theta}{2\mu}, \quad (\text{B13a})$$

$$b_{21}^0 = \frac{3}{2} + \frac{15}{4\mu} - \frac{9\theta}{4\mu}, \quad (\text{B13b})$$

$$b_{22}^0 = \frac{15}{4} + \frac{105}{16\mu} - \frac{45\theta}{16\mu}, \quad (\text{B13c})$$

$$b_{00}^1 = \frac{1}{2} + \frac{1}{2\mu}, \quad (\text{B14a})$$

$$b_{01}^1 = \frac{3}{4} + \frac{3}{4\mu}, \quad (\text{B14b})$$

$$b_{02}^1 = \frac{15}{16} + \frac{15}{16\mu}, \quad (\text{B14c})$$

$$b_{10}^1 = \frac{3}{4} + \frac{9}{4\mu} - \frac{3\theta}{2\mu}, \quad (\text{B15a})$$

$$b_{11}^1 = \frac{27}{8} + \frac{45}{8\mu} - \frac{9\theta}{4\mu}, \quad (\text{B15b})$$

$$b_{12}^1 = \frac{225}{32} + \frac{315}{32\mu} - \frac{45\theta}{16\mu}, \quad (\text{B15c})$$

$$b_{20}^1 = \frac{15}{16} + \frac{75}{16\mu} - \frac{15\theta}{4\mu}, \quad (\text{B16a})$$

$$b_{21}^1 = \frac{225}{32} + \frac{525}{32\mu} - \frac{75\theta}{8\mu}, \quad (\text{B16b})$$

$$b_{22}^1 = \frac{2625}{128} + \frac{4725}{128\mu} - \frac{525\theta}{32\mu}, \quad (\text{B16c})$$

$$b_{00}^2 = 1 + \frac{3}{2\mu} - \frac{\theta}{2\mu}, \quad (\text{B17a})$$

$$b_{01}^2 = 3 + \frac{15}{4\mu} - \frac{3\theta}{4\mu}, \quad (\text{B17b})$$

$$b_{02}^2 = \frac{45}{8} + \frac{105}{16\mu} - \frac{15\theta}{16\mu}, \quad (\text{B17c})$$

$$b_{10}^2 = 3 + \frac{15}{2\mu} - \frac{9\theta}{2\mu}, \quad (\text{B18a})$$

$$b_{11}^2 = 15 + \frac{105}{4\mu} - \frac{45\theta}{4\mu}, \quad (\text{B18b})$$

$$b_{12}^2 = \frac{315}{8} + \frac{945}{16\mu} - \frac{315\theta}{16\mu}, \quad (\text{B18c})$$

$$b_{20}^2 = \frac{45}{8} + \frac{315}{16\mu} - \frac{225\theta}{16\mu}, \quad (\text{B19a})$$

$$b_{21}^2 = \frac{315}{8} + \frac{2835}{32\mu} - \frac{1575\theta}{32\mu}, \quad (\text{B19b})$$

$$b_{22}^2 = \frac{8505}{64} + \frac{31185}{128\mu} - \frac{14175\theta}{128\mu}. \quad (\text{B19c})$$

APPENDIX C: SMALL MASS-RATIO APPROXIMATION

In a pure plasma with comparable temperatures of electrons and ions ($T_e \sim T_i, \theta \sim 1$), the electron-ion and ion-electron collision operators C_{ei} and C_{ie} can be significantly simplified by neglecting the higher orders of mass ratio of electrons to ions.

1. Electron-ion operator, $\mu \gg 1$

For $a=e, b=i$, we have $\mu \gg 1, \chi \ll 1, E'_b(v_{Ta}/v_{Tb}) \sim 0$, and $E_b(v_{Ta}/v_{Tb}) \sim 1$. Noting the factor of $\chi \sim \mu^{-1/2}$ in Eq. (46) and $\chi^{-1} \sim \mu^{1/2}$ in Eq. (48), we ignore lower orders than $O(\mu^0)$ in Eq. (47)

$$a_{E0ab}^{lm} \approx O(\mu^{-1}), \quad (\text{C1a})$$

$$a_{E1ab}^{lm} \approx -\frac{1}{2}l(l+1), \quad (\text{C1b})$$

$$a_{E2ab}^{lm} \approx O(\mu^{-1}), \quad (\text{C1c})$$

$$a_{e0ab}^{lm} \approx \frac{2}{\theta}(1-\theta), \quad (\text{C1d})$$

$$a_{e1ab}^{lm} \approx (l+2m), \quad (\text{C1e})$$

$$a_{e2ab}^{lm} \approx O(\mu^{-1}), \quad (\text{C1f})$$

and $O(\mu)$ in Eq. (49)

$$b_{e0ab}^{lm} \approx \frac{2\mu}{\theta^2}, \quad (\text{C2a})$$

$$b_{e1ab}^{lm} \approx O(\mu^0), \quad (\text{C2b})$$

$$b_{+0ab}^{lm} \approx \frac{4l\mu}{(2l+1)\theta}, \quad (\text{C2c})$$

$$b_{+1ab}^{lm} \approx O(\mu^0), \quad (C2d)$$

$$b_{-0ab}^{lm} \approx -\frac{4(l+1)\mu}{(2l+1)\theta}, \quad (C2e)$$

$$b_{-1ab}^{lm} \approx O(\mu^0). \quad (C2f)$$

a. Collision operators

Using

$$I_{+b}^{2k} \approx \frac{(k-1/2)!}{(-1/2)!}, \quad (C3)$$

$$I_{-b}^{2k+1} \approx 0 \quad (C4)$$

the leading orders comparable to the like-species collision operator are:

$$C_{ab}^{(lk,0)} \approx -\frac{l(l+1)}{2s_a^3} \nu_{ab} f_a^{(0)}(v) \mathbf{P}_a^{lk} \cdot \hat{\mathbf{M}}_a^{lk}, \quad (C5a)$$

$$C_{ab}^{(0,lk)} \approx \delta_{l1} \delta_{k0} \frac{\chi}{s_a^3} \nu_{ab} f_a^{(0)}(v) \mathbf{P}_a^{10} \cdot \hat{\mathbf{M}}_b^{10}. \quad (C5b)$$

b. Collision coefficients

From the approximations

$$I_{eab}^k \approx \frac{(k/2-1/2)!}{(-1/2)!}, \quad (C6)$$

$$I_{eba}^k = O(1/\sqrt{\mu}), \quad (C7)$$

$$I_{Eab}^{2k+1} = O(1/\sqrt{\mu}), \quad (C8)$$

$$I_{Eba}^{2k+1} \approx \frac{k!}{2}, \quad (C9)$$

we can write the leading orders:

$$\alpha_{Enab}^{lqm} \approx \frac{4n_a \nu_{ab}}{\pi^{1/2}} \frac{(l+q+m-n)!}{2}, \quad (C10)$$

$$\alpha_{enab}^{lqm} \approx \frac{4n_a \nu_{ab}}{\pi^{1/2}} O(\mu^{-1}). \quad (C11)$$

Then the collision coefficients A_{ab}^{lpk} are

$$A_{ab}^{lpk} \approx -\frac{n_a \nu_{ab}}{\pi^{1/2}} l(l+1) \sum_{m=0}^k c_{km}^l \sum_{q=0}^p c_{pq}^l (l+q+m-1)!, \quad (C12)$$

or,

$$A_{ab}^{lpk} \approx -\frac{n_a \nu_{ab}}{\pi^{1/2}} l(l+1) \sum_{m=0}^k c_{km}^l \frac{(l+m-1)!(p-m+1/2)!}{(1/2-m)!p!}. \quad (C13)$$

Similarly, the only dominant term in β is

$$\beta_{+nab}^{lqm} \approx \frac{4n_a \nu_{ab} \chi^l}{\sqrt{\pi} 2} (q+n)! \left(\frac{\theta}{\mu}\right)^{1-n} \frac{(l+m+1/2)!}{(-1/2)!} \quad (C14)$$

and the leading term is

$$B_{ab}^{lpk} \approx \frac{4n_a \nu_{ab} \chi}{\pi^{1/2}} \sum_{q=0}^p c_{pq}^1 q! \delta_{k0} \delta_{l1} \quad (C15)$$

or

$$B_{ab}^{lpk} \approx \frac{4n_a \nu_{ab}}{\pi^{1/2}} \chi \frac{(p+1/2)!}{\pi^{1/2} p!} \delta_{k0} \delta_{l1}. \quad (C16)$$

2. Ion-electron operator, $\mu \ll 1$

For $a=i, b=e$, from $\mu \ll 1, \chi \gg 1$ one can show that

$$C_{ie} \ll C_{ii} \quad (C17)$$

and hence the ion-electron collision coefficients are negligible:

$$A_{ie} \sim B_{ie} \ll A_{ii} \sim B_{ii}. \quad (C18)$$

Therefore, the ion species with electrons can be considered effectively as a single species.

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